

Chapter 6

Spectral properties

If a linear map has equal input and output space, like $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$, we can assign a *spectrum* to it. As outlined in Sec.1.6 the spectrum can be defined in purely algebraic terms as the set of complex numbers λ for which $(\lambda \text{id} - T)$ is not invertible. Since we restrict ourselves to finite dimensions this is equivalent to the set of λ 's for which there is an $X \in \mathcal{M}_d(\mathbb{C})$ such that

$$T(X) = \lambda X. \quad (6.1)$$

Although X is an operator, will refer to it as an *eigenvector* of T as $\mathcal{M}_d(\mathbb{C})$ is considered as a vector space in this context. Hence, the spectrum of T is the collection of eigenvalues of the $d^2 \times d^2$ matrix representation $\hat{T}$ introduced in Sec.2.3. Using this representation we see that the eigenvalues of T and T^* coincide due to $\hat{T}^* = \hat{T}^T$ when using a Hermitian basis of operators in Eq.(2.20).

Location of eigenvalues: For Hermiticity preserving maps (i.e., in particular for positive maps) the adjoint of Eq.(6.1) implies that eigenvalues are either real or they come in complex conjugate pairs.¹ Moreover, for every real λ there is at least one corresponding Hermitian eigenvector X .

Recall that the *spectral radius* is defined as $\varrho(T) := \sup\{|\lambda| \mid \lambda \in \text{spec}(T)\}$, i.e., as the eigenvalue which is largest in magnitude.

Proposition 6.1 (Spectral radius of positive maps) *If T is a positive map on $\mathcal{M}_d(\mathbb{C})$, then its spectral radius satisfies*

$$\varrho(T) \leq \|T(\mathbb{1})\|_\infty. \quad (6.2)$$

If in addition T is unital or trace-preserving, then there is an eigenvalue $\lambda = 1$. That is, in this case $\varrho(T) = 1$ so that all eigenvalues of T lie in the unit disc of the complex plane.

¹More precisely, if we decompose X into Hermitian and anti-Hermitian parts and use the Hermiticity preservation together with linearity we obtain $T(X^\dagger) = \bar{\lambda}X^\dagger$.

For a more detailed investigation of the spectral radius see the subsequent section (in particular Thm. 6.5).

PROOF By the theorem of Russo and Dye we have $\|T(X)\|_\infty \leq \|T(\mathbb{1})\|_\infty \|X\|_\infty$. Therefore if $T(X) = \lambda X$, then

$$|\lambda| \|X\|_\infty = \|T(X)\|_\infty \leq \|T(\mathbb{1})\|_\infty \|X\|_\infty, \quad (6.3)$$

which implies Eq.(6.2). If T is unital then $T(\mathbb{1}) = \mathbb{1}$ provides an eigenvalue $\lambda = 1$. Similar holds if T is trace-preserving, since then T^* (which has the same spectrum) is unital. $\square$

Another way to see that the spectral radius of every quantum channel (or trace-preserving positive map) is one would be to use that $\varrho(\hat{T}) = \lim_{n \rightarrow \infty} \|\hat{T}^n\|^{1/n}$ (see Eq.(1.35)) together with the fact that T^n remains a quantum channel for every n and that quantum channels in finite dimensions have bounded norm (see ...).

To summarize, the eigenvalues of every quantum channel are restricted to lie on the unit disc, there is one eigenvalue one and the others are real or come in complex conjugate pairs. In fact, there is no further restriction (as in the case of 'classical' stochastic matrices): for every $d > 1$, every $\lambda : |\lambda| \leq 1$ can appear as an eigenvalue of a quantum channel on $\mathcal{M}_d$. Only the set of eigenvalues as a whole underlies further non-trivial constraints (see Sec.6.1).

Spectral decomposition: Recall some basic linear algebra. Like every other square matrix, $\hat{T} \in \mathcal{M}_{d^2}(\mathbb{C})$ admits a *Jordan decomposition* of the form

$$\hat{T} = X \left(\bigoplus_{k=1}^K J_k(\lambda_k) \right) X^{-1}, \quad J_k(\lambda) := \begin{pmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \lambda & 1 \\ & & & \lambda \end{pmatrix} \in \mathcal{M}_{d_k}(\mathbb{C}), \quad (6.4)$$

where the J_k 's are *Jordan blocks* of size d_k (with $\sum_k d_k = d^2$) and the number K of Jordan blocks equals the number of different non-zero eigenvectors. It is sometimes useful to subdivide each Jordan block into a projection² and a nilpotent part so that we get

$$\begin{aligned} \hat{T} &= \sum_{k=1}^K \lambda_k P_k + N_k, & N_k^{d_k} &= 0, \quad N_k P_k = P_k N_k = N_k, \\ & & P_k P_l &= \delta_{kl} P_k, \quad \text{tr}[P_k] = d_k, \quad \sum_k P_k = \mathbb{1}. \end{aligned} \quad (6.5)$$

The number of Jordan blocks with eigenvalue λ is the *geometric multiplicity* of λ , while their joint dimension $\sum_{k:\lambda_k=\lambda} d_k$ is its *algebraic multiplicity*. If these two multiplicities are equal for every eigenvalue, then $\hat{T}$ is called *non-defective*. Non-defective matrices are dense (which is easily seen by perturbing the eigenvalues)

²This implies that in Eq.(6.5) $P_k^2 = P_k$ but not necessarily $P_k = P_k^\dagger$, i.e., P_k need not be an *orthogonal projection*.

and exactly those for which complete bases of eigenvectors exist. We can then write

$$\hat{T} = \sum_k \lambda_k |R_k\rangle\langle L_k|, \quad \langle L_k, R_l\rangle = \delta_{kl}, \quad (6.6)$$

so that the right eigenvectors $\{R_k\}$ and the left eigenvectors $\{L_k\}$ are biorthogonal.

Returning to the fact that T is a linear map on $\mathcal{M}_d(\mathbb{C})$ this means that in the non-defective case there are biorthogonal operator bases $\{R_k \in \mathcal{M}_d(\mathbb{C})\}$ and $\{L_k \in \mathcal{M}_d(\mathbb{C})\}$ such that

$$T(A) = \sum_k \lambda_k \operatorname{tr}[L_k^\dagger A] R_k, \quad \operatorname{tr}[L_k^\dagger R_l] = \delta_{kl}, \quad (6.7)$$

$$\text{so that } \forall k : T(R_k) = \lambda_k R_k. \quad (6.8)$$

The Choi-Jamiolkowski operator $\tau = (T \otimes \operatorname{id})(|\Omega\rangle\langle\Omega|)$ then takes the form

$$\tau = \frac{1}{d} \sum_k \lambda_k \bar{L}_k \otimes R_k. \quad (6.9)$$

If T is a trace-preserving map, then all right eigenvectors R_k corresponding to eigenvalues $\lambda_k \neq 1$ have to be traceless, $\operatorname{tr}[R_k] = 0$ since $\mathbb{1}$ is a left eigenvector.

Note that if T is a quantum channel, then $\hat{T}$ is generally neither Hermitian (unless $T = T^*$, see Prop.2.6) nor is it necessarily non-defective. However, the uniformly bounded norm of trace-preserving positive maps prevents eigenvalues of magnitude one (the so-called *peripheral spectrum*) from having non-trivial Jordan blocks (meaning $d_k > 1$):

Proposition 6.2 (Trivial Jordan blocks for peripheral spectrum) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving (or unital) positive linear map. If λ is an eigenvalue of T with $|\lambda| = 1$, then its geometric multiplicity equals its algebraic multiplicity, i.e., all Jordan blocks for λ are one-dimensional.*

PROOF First note that for every trace-preserving (or unital) positive map $\operatorname{tr}[AT(B)]$ can be upper bounded in terms of the dimension d and the norms of A and B only. While we will discuss such bounds in greater detail later it suffices for now to note that we may write A and B as a linear combination of positive matrices and that for positive A, B we have $\operatorname{tr}[AT(B)] \leq \|A\|_\infty \|B\|_\infty \operatorname{tr}[\mathbb{1}T(\mathbb{1})] = d\|A\|_\infty \|B\|_\infty$. The same bound has to hold for every power T^n , $n \in \mathbb{N}$. However, if we take the n 'th power of a Jordan block $J(\lambda)$, it is a Toeplitz matrix whose first row reads

$$[J(\lambda)^n]_{1j} = \lambda^{n-j+1} \binom{n}{j-1}, \quad j = 1, \dots, \dim(J(\lambda)). \quad (6.10)$$

Hence, if $\dim(J(\lambda)) > 1$ and $|\lambda| = 1$ this would contain entries which grow unboundedly with n contradicting the uniform boundedness of T^n . $\square$

We can now use the obtained results in order to show that some of the spectral projections appearing in Eq.(6.5) are (completely) positive in their own right. Define

$$\hat{T}_\infty := \sum_{k:\lambda_k=1} P_k, \quad (6.11)$$

$$\hat{T}_\phi := \sum_{k:|\lambda_k|=1} P_k, \quad (6.12)$$

$$\hat{T}_\varphi := \sum_{k:|\lambda_k|=1} \lambda_k P_k, \quad (6.13)$$

from the spectral decomposition in Eq.(6.5) and the corresponding maps $T_\infty, T_\phi, T_\varphi$ via Eq.(2.20).

Proposition 6.3 (Cesaro means) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a linear map which is trace-preserving and (completely) positive. Then $T_\infty, T_\phi, T_\varphi$ are trace preserving and (completely) positive as well. More precisely, we have that (i) there exists an increasing sequence $n_i \in \mathbb{N}$ such that $\lim_{i \rightarrow \infty} T^{n_i} = T_\phi$, (ii) $T_\varphi = TT_\phi$ and (iii)*

$$T_\infty = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N T^n. \quad (6.14)$$

PROOF Showing (i)-(iii) essentially proves the claim that the three maps are trace-preserving and (completely) positive. Throughout we will use that the spectral radius of T is one, that there is an eigenvalue one and that the peripheral spectrum of T only has one-dimensional Jordan blocks. For (i) we exploit the subsequent Lemma on simultaneous Diophantine approximations. For our purpose Lemma 6.1 implies that for every $\epsilon > 0$ there is an $n \in \mathbb{N}$ such that $|\lambda_k^n - 1| \leq \epsilon$ for all eigenvalues λ_k of magnitude one. This means that there is an increasing sequence $n_i \in \mathbb{N}$ such that T^{n_i} is a better and better approximation to T_ϕ . Since the set of (completely) positive, trace-preserving maps on $\mathcal{M}_d$ is compact, the limit point T_ϕ belongs to this set. (ii) is obvious from the spectral decomposition in Eq.(6.5) and (iii) is an immediate consequence of the geometric series $\sum_{n=1}^N \lambda^n = (\lambda - \lambda^{N+1})/(1 - \lambda)$ for $\lambda \neq 1$. $\square$

In a similar vein as T_∞ is expressed as a *Cesaro mean* in Eq.(6.14) we can write

$$T_\phi = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{k:|\lambda_k|=1} (\lambda_k T)^n. \quad (6.15)$$

The maps T_∞ and T_ϕ are projections onto the set of fixed points (see Chap.6.4) and the space spanned by the eigenvectors of the peripheral spectrum respectively. That is, $X \in \mathcal{M}_d$ is in the range of T_∞ iff $T(X) = X$ and it is in the range of T_ϕ iff for every $\epsilon > 0$ there is an $n \in \mathbb{N}$ such that $\|T^n(X) - X\| \leq \epsilon$. In other words the image of T_ϕ is the set of operators for which recurrences arise.

For completeness (and later use) Dirichlet's theorem on simultaneous Diophantine approximations which was used in the above proof:

Lemma 6.1 (Dirichlet's theorem) *Let $x_1, \dots, x_m$ be m real numbers and $q > 1$ any integer. Then there exist integers $n, p_1, \dots, p_m$ such that*

$$1 \leq n \leq q^m \quad \text{and} \quad |x_k n - p_k| \leq \frac{1}{q}, \quad \forall k. \quad (6.16)$$

Resolvents: The *resolvent* R of a linear operator T is an operator-valued function on $\mathbb{C}$ defined as

$$R(z) := (z \text{id} - T)^{-1}. \quad (6.17)$$

The singularities of R are exactly the eigenvalues of T . Hence, R is *meromorphic* and defined on the set which is the complement of the spectrum of T , called *resolvent set*. For $|z| > \varrho(T)$, i.e., outside the spectral radius of T , we can express the resolvent as

$$R(z) = \frac{1}{z} \sum_{t=0}^{\infty} \frac{T^t}{z^t}. \quad (6.18)$$

If T^t describes the discrete time evolution of a quantum (or classical) system this series implies that the spectrum of T is reflected in a peculiar way in the time evolution of expectation values: consider any $\rho, A \in \mathcal{M}_d(\mathbb{C})$ and the sequence $\langle A(t) \rangle := \text{tr}[\rho T^t(A)]$ for $t = 0, 1, 2, \dots$. If A represents an observable and ρ is a density matrix evolving according to T , then $\langle A(t) \rangle$ is the expectation value at time t . The *z-transform* (or *discrete Laplace transform*) of the 'signal' $\langle A(t) \rangle$ is given by

$$\mathcal{L}(z) := \frac{1}{z} \sum_{t \in \mathbb{N}_0} \frac{\langle A(t) \rangle}{z^t}, \quad (6.19)$$

which converges for $|z| > \varrho(T)$ and can be defined in the interior of that disc by analytic continuation. Obviously, the poles of $\mathcal{L}$ are located at eigenvalues of T . Note, however, that depending on A and ρ not every eigenvalue of T will give rise to a pole of $\mathcal{L}$.

Using the resolvent, many of the results of analytic function theory carry over to the world of operators. For instance if $\Delta \subseteq \mathbb{C}$ is simply connected and enclosed by a smooth curve $\partial\Delta$, then

$$\sum_{k: \lambda_k \in \Delta} P_k = \frac{1}{2\pi i} \int_{\partial\Delta} R(z) dz, \quad \text{and} \quad (6.20)$$

$$f(T) = \frac{1}{2\pi i} \int_{\partial\Delta} f(z) R(z) dz, \quad \text{if } \lambda_k \in \Delta \quad \forall k, \quad (6.21)$$

where $f : \mathbb{C} \rightarrow \mathbb{C}$ is any function holomorphic in Δ and the P_k 's are the projections appearing in the spectral decomposition in Eq.(6.5).

6.1 Determinants

The determinant (=product of all eigenvalues) is a useful tool in particular when it comes to the composition of maps since

$$\det(T_1 T_2) = \det(T_1) \det(T_2). \quad (6.22)$$

In fact, the property of being multiplicative essentially characterizes the determinant: if a functional $F : \mathcal{M}_D(\mathbb{C}) \rightarrow \mathbb{C}$ satisfies $F(T_1 T_2) = F(T_1)F(T_2)$ for all $T_i \in \mathcal{M}_D(\mathbb{C})$, then $F(T) = f(\det(T))$ for some multiplicative function $f : \mathbb{C} \rightarrow \mathbb{C}$ [?].

Also note that Eq.(6.22) implies the simple relation $\prod_i s_i = |\det(T)|$ between the determinant and the singular values $\{s_i\}$ of a map.

The following specifies the possible range of determinants for positive and trace-preserving maps on $\mathcal{M}_d(\mathbb{C})$ and characterizes the extremal values:

Theorem 6.1 (Determinants) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_d$ be a positive and trace preserving linear map.*

1. $\det T$ is real and contained in the interval $[-1, 1]$,
2. $|\det T| = 1$ iff T is either a unitary conjugation (i.e., $T(A) = UAU^\dagger$ for some unitary $U \in \mathcal{M}_d$) or unitarily equivalent to a matrix transposition (i.e., $T(A) = UA^T U^\dagger$ for some unitary $U \in \mathcal{M}_d$),
3. if T is a unitary conjugation then $\det T = 1$ and if $\det T = -1$ then T is a matrix transposition up to unitary equivalence. In both cases the converse holds iff $\lfloor \frac{d}{2} \rfloor$ is odd.

PROOF Since eigenvalues of positive maps either come in complex conjugate pairs or are real, their product $\det T$ is real, too. Due to the additional trace-preserving property the spectral radius is one (Prop.6.1) so that $\det T \in [-1, 1]$.

Now consider the case $\det T = \pm 1$ where all eigenvalues are phases. Following Dirichlet's Theorem (Lem.6.1) as in the first part of Prop.6.3, there is always a subsequence n_i such that the limit of powers $\lim_{i \rightarrow \infty} T^{n_i} =: T_\phi$ has eigenvalues which all converge to one which implies that $T_\phi = \text{id}$. Hence, the inverse $T^{-1} = T_\phi T^{-1} = \lim_{i \rightarrow \infty} T^{n_i-1}$ is a trace-preserving positive map as well.

Assume that the image of any pure state density matrix Ψ under T is mixed, i.e., $T(\Psi) = \lambda \rho_1 + (1 - \lambda) \rho_2$ with $\rho_1 \neq \rho_2$, $\lambda \in (0, 1)$. Then by applying T^{-1} to this decomposition we would get a nontrivial convex decomposition for Ψ (due to positivity of T^{-1}) leading to a contradiction. Hence, T and its inverse map pure states onto pure states. Furthermore, they are unital, which can again be seen by contradiction. So assume $T(\mathbb{1}) \neq \mathbb{1}$. Then the smallest eigenvalue of $T(\mathbb{1})$ satisfies $\lambda_{\min} < 1$ due to the trace preserving property. If we denote by $|\lambda\rangle$ a corresponding eigenvector, then $\mathbb{1} - \frac{\lambda_{\min} + 1}{2} T^{-1}(|\lambda\rangle\langle\lambda|)$ is a positive operator, but its image under T would no longer be positive. Therefore we must have $T(\mathbb{1}) = \mathbb{1}$.

Every unital positive trace preserving map is contractive with respect to the Hilbert-Schmidt norm. As this holds for both T and T^{-1} we have that $\forall A \in \mathcal{M}_d : \|T(A)\|_2 = \|A\|_2$, i.e., T acts unitarily on the Hilbert-Schmidt Hilbert space. In particular, it preserves the Hilbert Schmidt scalar product $\text{tr}[T(A)T(B)^\dagger] = \text{tr}[AB^\dagger]$. Applying this to pure states $A = |\phi\rangle\langle\phi|$ and $B = |\psi\rangle\langle\psi|$ shows that T gives rise to a mapping of the Hilbert space onto itself which preserves the value of $|\langle\phi|\psi\rangle|$. By Wigner's theorem (Thm. 1.1) this has to be either unitary or anti-unitary. If T is a unitary conjugation then $\det T = \det(U \otimes \bar{U}) = 1$ due to Eq.(2.21). Since every anti-unitary is unitarily equivalent to complex conjugation, we get that T is in this case a matrix transposition $T(A) = A^T$ (up to unitary equivalence). The determinant of the matrix transposition is easily seen in the Gell-Mann basis of $\mathcal{M}_d$ (see Hermitian operator bases in Sec.2.3). That is, we take basis elements F_α of the form $\sigma_x/\sqrt{2}, \sigma_y/\sqrt{2}$ for $\alpha = 1, \dots, d^2 - d$ and diagonal for $\alpha = d^2 - d + 1, \dots, d^2$. In this basis matrix transposition is diagonal and has eigenvalues 1 and -1 where the latter appears with multiplicity $d(d-1)/2$. This means that matrix transposition has determinant minus one iff $d(d-1)/2$ is odd, which is equivalent to $\lfloor \frac{d}{2} \rfloor$ being odd. $\square$

From this we get the following corollaries:

Corollary 6.1 (Monotonicity of the determinant) *Let T, T' be two positive and trace-preserving linear maps on $\mathcal{M}_d$. Then the determinant is decreasing in magnitude under composition, i.e., $|\det T| \geq |\det TT'|$ where equality holds iff T' is a unitary conjugation, a matrix transposition or $\det T = 0$.*

Corollary 6.2 (Positive invertible maps) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_d$ be a trace-preserving and positive linear map. Then the inverse T^{-1} is positive, too iff T is a unitary conjugation or matrix transposition.*

Note that the inverse (if it exists) of a trace-preserving linear map is automatically trace-preserving. Hence, the above statement implies that the inverse of a quantum channel is again a quantum channel iff it describes a unitary time evolution.

One might wonder whether completely positive maps can have negative determinants. The following simple example answers this question in the affirmative. It is build up on the map $\rho \mapsto \rho^{T_c}$ which transposes the corners of $\rho \in \mathcal{M}_d$, i.e., $(\rho^{T_c})_{k,l}$ is $\rho_{l,k}$ for the entries $(k,l) = (1,d), (d,1)$ and remains $\rho_{k,l}$ otherwise. Note that for $d = 2$ this is the ordinary matrix transposition.

Example 6.1 (Negative determinant) *The map $T : \mathcal{M}_d \rightarrow \mathcal{M}_d$ defined by*

$$T(\rho) = \frac{\rho^{T_c} + \mathbb{1} \text{tr}[\rho]}{1+d} \quad (6.23)$$

is trace preserving, completely positive with Kraus rank $d^2 - 1$ and has determinant $\det T = -(d+1)^{1-d^2}$. For $d = 2$ the channel is entanglement breaking and can be written as

$$T(\rho) = \frac{1}{3} \sum_{j=1}^6 |\bar{\xi}_j\rangle\langle\xi_j| \rho |\xi_j\rangle\langle\bar{\xi}_j|, \quad (6.24)$$

where the six ξ_j are the normalized eigenvectors of the three Pauli matrices. Moreover, the example given in Eq.(6.24) minimizes the determinant within the set of all quantum channels on $\mathcal{M}_2$.

PROOF A convenient matrix representation of the channel is given in the generalized Gell-Mann basis. Choose F_1 as the $\sigma_y/\sqrt{2}$ element corresponding to the corners and $F_2 = \mathbb{1}/\sqrt{d}$ the only traceless element. Then $\hat{T} = \text{diag}[-1, 1 + d, 1, \dots, 1]/(d + 1)$ leading to $\det T = -(d + 1)^{1-d^2}$.

For complete positivity we have to check positivity of the Jamiolkowski state τ . The corner transposition applied to a maximally entangled state leads to one negative eigenvalue $-1/d$. This is, however, exactly compensated by the second part of the map such that $\tau \geq 0$ with rank $d^2 - 1$.

The representation for $d = 2$ is obtained from $\text{tr}[AT(B)] = \text{tr}[(A \otimes B^T)\tau]/d$ by noting that in this case τ is proportional to the projector onto the symmetric subspace which in turn can be written as $\frac{1}{2} \sum_j |\xi_j\rangle\langle\xi_j|^{\otimes 2}$ in agreement with the given Kraus representation of the channel.

In order to see that the determinant is minimized for $d = 2$, recall from Sec.2.4 that any trace-preserving T can be conveniently represented in terms of the real 4×4 matrix $\hat{T}_{ij} := \text{tr}[\sigma_i T(\sigma_j)]/2$ where the σ_i s are identity and Pauli matrices:

$$\hat{T} = \begin{pmatrix} 1 & 0 \\ v & \Lambda \end{pmatrix}, \quad (6.25)$$

where $v \in \mathbb{R}^3$, Λ is a 3×3 matrix and $\det \Lambda = \det T$. To simplify matters we can diagonalize Λ by special orthogonal matrices $O_1 \Lambda O_2 = \text{diag}\{\lambda_1, \lambda_2, \lambda_3\}$ corresponding to unitary operations before and after the channel (see Exp.1.1 for the relation between $SO(3)$ and $SU(2)$). Obviously, this does neither change the determinant, nor complete positivity. For the latter it is necessary that $\vec{\lambda}$ is contained in a tetrahedron spanned by the four corners of the unit cube with $\lambda_1 \lambda_2 \lambda_3 = 1$. Fortunately, all these points can indeed be reached by unital channels ($v = 0$) for which this criterion becomes also sufficient for complete positivity. By symmetry we can restrict our attention to one octant and reduce the problem to maximizing $p_1 p_2 p_3$ over all probability vectors $\vec{p}$ yielding $\vec{p} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}) = (\lambda_1, -\lambda_2, \lambda_3)$. Hence, the minimal determinant is $-(\frac{1}{3})^3$ and the corresponding channel can easily be constructed from $\hat{T}$ as $T : \rho \mapsto \frac{1}{3}(\rho^T + \mathbb{1})$. $\square$

The example of a completely positive map with negative determinant constructed above has at least Kraus rank 3 (for $d = 2$). In fact, this is the minimal number of Kraus operators which allows for a negative determinant:

Proposition 6.4 (Positive determinant for small Kraus rank) *If $T : \mathcal{M}_d \rightarrow \mathcal{M}_d$ is a completely positive linear map with Kraus rank at most two, then $\det T \geq 0$.*

PROOF If $A, B \in \mathcal{M}_d$ are Kraus operators then $\hat{T} = A \otimes \bar{A} + B \otimes \bar{B}$ is a matrix representation of T (see Eq.(2.21)). Assume for the moment that $\det A \neq 0$. Then due to multiplicativity of the determinant

$$\det T = \det(A \otimes \bar{A}) \det(\mathbb{1} + A^{-1}B \otimes \bar{A}^{-1}\bar{B}) \geq 0. \quad (6.26)$$

In order to understand the inequality note that if $\{\lambda_j\}$ are the eigenvalues of $A^{-1}B$, then $\{1 + \lambda_i \bar{\lambda}_j\}$ are those of $\mathbb{1} + A^{-1}B \otimes \bar{A}^{-1}\bar{B}$. That is, they all

come in complex conjugate pairs so that their product is positive. The same argumentation holds, of course, for the $A \otimes \bar{A}$ term. The proof is completed by using that the set of maps with $\det A \neq 0$ is dense and that $\det T$ is continuous. $\square$

Thm. 6.1 shows that if for a quantum channel $\det T = 1$, then it has to be a unitary or, expressed in terms of the purity of the Jamiolkowsky state, $\text{tr} [\tau^2] = 1$. By continuity a channel close to a unitary, described by a τ with purity close to one, will still have determinant close to one. The following is a quantitative version of the converse: if the determinant is large, then the purity has to be large, too. We will see later that and how this implies closeness to a unitary for quantum channels.

Proposition 6.5 (Determinant and Choi-Jamiolkowski operator) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_d$ be a linear map and τ the corresponding Choi-Jamiolkowski operator. Then*

$$|\det T| \leq \text{tr} [\tau^\dagger \tau]^{d^2/2}. \quad (6.27)$$

PROOF To relate the 'purity' to the determinant we exploit that by Eq.(2.22) $\hat{T} = d\tau^\Gamma$ is a matrix representation of the map T , so that

$$\text{tr} [\tau^\dagger \tau] = \frac{1}{d^2} \text{tr} [\hat{T}^\dagger \hat{T}] = \frac{1}{d^2} \sum_{i=1}^{d^2} s_i^2, \quad (6.28)$$

where the s_i are the singular values of $\hat{T}$ (and thus T). From this Eq.(6.27) is obtained via the geometric–arithmetic mean inequality together with the fact that $|\det T| = \prod_i s_i$. $\square$

6.2 Irreducible maps and Perron-Frobenius theory

Perron-Frobenius theory, roughly speaking, deals with the dominant eigenvalues and eigenvectors of (element-wise) non-negative matrices (see Exp. ...), i.e., in particular with those of classical channels. In order to make the theory work the considered maps have to fulfill a generic condition like *irreducibility*, *primitivity* or *strict positivity* (in order of increasing restriction).

In this section we will discuss the generalization of Perron-Frobenius theory to the context of positive maps on $\mathcal{M}_d(\mathbb{C})$. The following theorem defines *irreducible* positive maps and shows that this can be done in various equivalent ways:

Theorem 6.2 (Irreducible positive maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive linear map. The following properties are equivalent:*

1. Irreducibility: *If $P \in \mathcal{M}_d(\mathbb{C})$ is a Hermitian projector such that $T(P\mathcal{M}_dP) \subseteq P\mathcal{M}_dP$, then $P \in \{0, \mathbb{1}\}$.*

2. For every non-zero $A \geq 0$ we have $(\text{id} + T)^{d-1}(A) > 0$.
3. For every non-zero $A \geq 0$ and every strictly positive $t \in \mathbb{R}$ we have $\exp[tT](A) > 0$.
4. For every orthogonal pair of non-zero, positive semi-definite $A, B \in \mathcal{M}_d(\mathbb{C})$ there is an integer $t \in \{1, \dots, d-1\}$ such that $\text{tr}[BT^t(A)] > 0$.

PROOF 1. $\rightarrow$ 2.: From $T(A) \geq 0$ we get an inclusion for the kernels $\ker(\text{id} + T)(A) \subseteq \ker A$. Suppose equality holds in this inclusion, then $\text{supp} T(A) \subseteq \text{supp} A$. Therefore $T(P\mathcal{M}_d P) \subseteq P\mathcal{M}_d P$ if we take P the Hermitian projector onto the support space of A . By 1. (irreducibility) this can only be if $A > 0$. Application of $(\text{id} + T)$ thus has to increase the rank until there is no kernel left, which happens after $d-1$ steps at the latest.

2. $\rightarrow$ 3.: Looking at the series expansion of both $(\text{id} + T)^{d-1}(A)$ and $\exp[tT](A)$ in terms of powers of T we realize that (i) all terms are positive, and (ii) all terms appearing in the first expansion are also present in the second one. Therefore we can bound $\exp[tT](A) \geq c(\text{id} + T)^{d-1}(A)$ with some non-zero positive constant c .

3. $\rightarrow$ 1.: Suppose T is reducible, i.e., there is a proper projection $P \neq 0, \mathbb{1}$ for which $T(P) \leq cP$ for some scalar $c > 0$. Then $\exp[tT](P) \leq \exp[tc]P$, so T does not fulfill 3..

4. $\rightarrow$ 1.: If T is reducible by a proper projection P , then $\text{tr}[(\mathbb{1} - P)T^t(P)] = 0$ for all $t \in \mathbb{N}$.

2. $\rightarrow$ 4.: Expanding $\text{tr}[B(\text{id} + T)^{d-1}(A)] > 0$ and using $\text{tr}[BA] = 0$ together with positivity of all terms in the expansion, we get that $\text{tr}[BT^t(A)] > 0$ for at least one $t \leq d-1$. $\square$

Note that T is irreducible iff the dual map T^* is irreducible. If a proper projection P 'reduces' T in the sense that $\text{tr}[(\mathbb{1} - P)T(P)] = 0$, then $\mathbb{1} - P$ reduces T^* .

In order to relate *irreducibility* to spectral properties of a positive map T it is useful to consider the following functionals defined on the cone of positive semi-definite operators:

$$r(X) := \sup\{\lambda \in \mathbb{R} \mid (T - \lambda \text{id})(X) \geq 0\}, \quad (6.29)$$

$$\tilde{r}(X) := \inf\{\lambda \in \mathbb{R} \mid (T - \lambda \text{id})(X) \leq 0\}. \quad (6.30)$$

We are especially interested in the maxima $r := \sup_{X \geq 0} r(X)$ and $\tilde{r} := \sup_{X \geq 0} \tilde{r}(X)$ which obviously satisfy $r \geq \tilde{r}$ and actually coincide for irreducible maps:

Theorem 6.3 (Spectral radius of irreducible maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be an irreducible positive map. Then*

1. For the quantities introduced below Eqs.(6.29,6.30) $r = \tilde{r}$.
2. r is a non-degenerate eigenvalue of T and the corresponding eigenvector is strictly positive, i.e., $T(X) = rX > 0$.

3. If there is any $\lambda > 0$ which is an eigenvalue of T with positive eigenvector, i.e., $T(Y) = \lambda Y \geq 0$, then $\lambda = r$.
4. r is the spectral radius of T .

PROOF First note that all suprema/infima in Eqs.(6.29,6.30) and for defining $r, \tilde{r}$ are attained since the respective sets are compact or can be made so, e.g., by imposing $\text{tr}[X] = 1$.

We begin with showing that r is attained for a strictly positive X and that $T(X) = rX$. To this end consider any non-zero $X \geq 0$ for which $\lambda := r(X) > 0$. The identity

$$(T + \text{id})^{d-1}(T - \lambda \text{id})(X) = (T - \lambda \text{id})(T + \text{id})^{d-1}(X), \quad (6.31)$$

shows two things. First, the supremum $\sup_{X \geq 0} r(X)$ is attained for a strictly positive X , since by Thm.6.2 $(T + \text{id})^{d-1}(X) > 0$. Second, the X achieving $r(X) = r$ must satisfy $(T - r \text{id})(X) = 0$, since otherwise the expression in Eq.(6.31) would be positive definite so that a larger multiple of the identity map could be subtracted—contradicting the supposed maximality. As $r(X) = \tilde{r}(X)$ for every eigenvector $X \geq 0$, the just mentioned observation together with the inequality $r \geq \tilde{r}$ proves item 1. of the theorem.

To prove 2. we still need to show non-degeneracy. Suppose there would be an eigenvector X' corresponding to r which is not a multiple of X . By taking the Hermitian (or i times the anti-Hermitian) part we can assume $X' = X'^\dagger$. As $X > 0$ there is a non-zero $c \in \mathbb{R}$ such that $0 \leq X + cX'$ has a kernel. This is, however, in conflict with

$$0 < (T + \text{id})^{d-1}(X + cX') = (r + 1)^{d-1}(X + cX'), \quad (6.32)$$

so that X' must be a multiple of X . Hence, r is indeed non-degenerate.

For the proof of 3. assume that $Y \geq 0$ is an eigenvector of T corresponding to an eigenvalue $\lambda > 0$. Using that r is an eigenvalue of the dual map T^* with a strictly positive eigenvector $X' > 0$ we get

$$r \text{tr}[X'Y] = \text{tr}[T^*(X')Y] = \text{tr}[X'T(Y)] = \lambda \text{tr}[X'Y]. \quad (6.33)$$

Since $\text{tr}[X'Y] > 0$ this implies $r = \lambda$.

Finally, in order to prove 4. we define a positive map $T'(\cdot) := X^{-1/2}T(X^{1/2} \cdot X^{1/2})X^{-1/2}/r$ with $X > 0$ being the positive eigenvector corresponding to the eigenvalue r of T . Note that up to the factor $1/r$ the maps T and T' differ only by a similarity transformation, i.e., their spectra coincide up to that factor. In particular the spectral radii satisfy $\varrho(T') = \varrho(T)/r$. Since T' is a unital positive map we have $\varrho(T') = 1$ by Prop.6.1 and thus $\varrho(T) = r$. $\square$

The proof of the last item of Thm.6.3 leads to the following simple observation:

Proposition 6.6 (Similarity transformations preserving irreducibility)
Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be an irreducible positive map. Then for arbitrary

constants $c > 0$ and invertible matrices $C \in \mathcal{M}_d(\mathbb{C})$ the maps

$$T'(\cdot) := c C^{-1} T(C \cdot C^\dagger) C^{-\dagger}, \quad (6.34)$$

are irreducible as well.

PROOF Assume that a projection Q would 'reduce' T' . Then the projection P onto the support of $CQC^\dagger$ would reduce T . By invertibility of C we have that P and Q have the same rank and thus T' can only be reducible if T is. $\square$

The invariance of irreducibility under similarity transformation of the form in Eq.6.34 shows that irreducibility does not only imply spectral properties of a positive map, but that it is determined by such properties:

Theorem 6.4 (Irreducibility from spectral properties) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive map with spectral radius ϱ . The following are equivalent:*

1. T is irreducible.
2. The spectral radius ϱ is a non-degenerate eigenvalue and the corresponding right and left eigenvectors are positive definite (i.e., $T(X) = \varrho X > 0$ and $T^*(Y) = \varrho Y > 0$).

PROOF 1. $\rightarrow$ 2. is given by Thm.6.3 when applied to T and T^* .

For the converse 2. $\rightarrow$ 1. we apply Eq.(6.34) with $c = 1/\varrho$ and $C = Y^{-1/2}$. In this way T' becomes trace preserving with $T(X') = X' > 0$, $X' := \sqrt{Y}X\sqrt{Y}$. Moreover, the non-degeneracy translates to that of the eigenvalue 1 of T' . Now assume that T' (and thus T) would be reducible, i.e., there is a Hermitian projection $P \notin \{0, \mathbb{1}\}$ such that $T' : \mathcal{A} \rightarrow \mathcal{A}$ with $\mathcal{A} := P\mathcal{M}_d(\mathbb{C})P$. In this case T' would have a fixed point density operator $\rho \in \mathcal{A}$ for which $T'(\rho) = \rho \geq 0$ (see Sec.6.4). Hence, the eigenvalue 1 of T would have to be degenerate since ρ is not a multiple of X' because of $X' \notin \mathcal{A}$ (as $X' > 0$). $\square$

If we consider the time evolution of a density operator of the form $\rho(t) := T^t(\rho)$ where T is a positive and trace-preserving map and $t \in \mathbb{N}$, then the previous theorem allows us to relate irreducibility to a unique time-average corresponding to a full rank stationary state:

Corollary 6.3 (Time-average and ergodicity) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive, trace-preserving linear map. Then the following are equivalent:*

1. T is irreducible,
2. There is a unique state with density matrix $\sigma > 0$ such that for every density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ we have

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N T^t(\rho) = \sigma. \quad (6.35)$$

PROOF Note that Eq.(6.35) is nothing but $T_\infty(\rho) = \sigma$, where T_∞ is the projection onto the eigenspace of T with eigenvalue one as given in Eq.(6.14). The equivalence 1. $\leftrightarrow$ 2. thus follows from the equivalence in Thm.6.4. $\square$

Irreducible positive maps are generic, i.e., in particular dense within the set of all positive maps. For instance $X \mapsto T(X) + \epsilon \mathbb{1} \text{tr}[X]$ is irreducible for every $\epsilon > 0$ and positive T since it maps every positive input to a positive definite output. By continuity of eigenvalues and eigenvectors we can therefore extend the above observation about the spectral radius to arbitrary positive maps:

Theorem 6.5 (Spectral radius and positive eigenvectors) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive map with spectral radius ϱ . Then ϱ is an eigenvalue and there is a positive semi-definite $X \in \mathcal{M}_d(\mathbb{C})$ such that $T(X) = \varrho X$.*

Of course, in general the spectral radius might be a degenerate eigenvalue and the corresponding positive eigenvector may have a kernel. Even for irreducible maps there can be several eigenvalues with modulus equal to the spectral radius:

The peripheral spectrum is the set of eigenvalues whose magnitude is equal to the spectral radius, i.e., equal to one if we are considering unital positive maps (Prop.6.1). As we saw in Prop.6.2 the properties positivity and unitality together imply that all eigenvalues in the peripheral spectrum have trivial (i.e., one-dimensional) Jordan blocks. If we add irreducibility and the assumed validity of the Schwarz inequality in Eq.(5.2) we can say much more:

Theorem 6.6 (Peripheral spectrum of irreducible Schwarz maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be an irreducible positive unital map which fulfills the Schwarz inequality in Eq.(5.2). Denote by $S := \text{spec}(T) \cap \exp(i\mathbb{R})$ the peripheral spectrum of T . Then*

1. *There is an integer $m \in \{1, \dots, d^2\}$ such that $S = \{\exp(2\pi i k/m)\}_{k \in \mathbb{Z}_m}$.*
2. *All eigenvalues in S are non-degenerate.*
3. *There is a unitary U such that for all $k \in \mathbb{N}$: $T(U^k) = \gamma^k U^k$ where $\gamma := \exp(2\pi i/m)$.*
4. *U has spectral decomposition $U = \sum_{k \in \mathbb{Z}_m} \gamma^k P_k$ where the spectral projections P_k satisfy $T(P_{k+1}) = P_k$.*

PROOF Take any eigenvalue $\lambda \in S$ with corresponding eigenvector $U \in \mathcal{M}_d(\mathbb{C})$, i.e., $T(U) = \lambda U$. Then the Schwarz inequality reads $U^\dagger U \leq T(U^\dagger U)$. Since T is irreducible by assumption we can apply the tools in the proof of Thm.6.3, in particular that $r(U^\dagger U) = 1$, which implies that equality has to hold in the Schwarz inequality and that U can be rescaled to be a unitary. Moreover, by Thm.5.3 we have

$$\bar{\lambda} U^\dagger T(X) = T(U^\dagger X), \quad \forall X \in \mathcal{M}_d(\mathbb{C}). \quad (6.36)$$

Applying this to the case where X is another eigenvector of T w.r.t. an eigenvalue in S shows that S is closed under multiplication and thus a group. Since there are at most d^2 elements in S we must have $S = \exp(2\pi i \mathbb{Z}_m/m)$ for some integer $m \leq d^2$.

Now assume that U and V are both eigenvectors w.r.t. the same eigenvalue in S . Then $T(U^\dagger V) = U^\dagger V$ and, since the eigenvalue one of an irreducible unital positive map is non-degenerate (Prop.6.1 and Thm.6.3), we must have $U \propto V$. Hence, all eigenvalues in S are simple.

If U is the unitary eigenvector corresponding to the eigenvalue $\gamma := \exp(2\pi i/m)$, then the group property gives $T(U^k) = \gamma^k U^k$ for all $k \in \mathbb{N}$. In particular $T(U^m) = U^m$ which has to be equal to $\mathbb{1}$ by non-degeneracy of the eigenvalue one. Therefore we can write $U = \sum_{k \in \mathbb{Z}_m} \gamma^k P_k$ with orthogonal spectral projections P_k . This gives

$$\sum_{k \in \mathbb{Z}_m} \gamma^{k+1} P_k = \gamma U = T(U) = \sum_{k \in \mathbb{Z}_m} \gamma^k T(P_k). \quad (6.37)$$

Due to $T(U^k) = \gamma^k U^k$, T acts as an automorphism on the $*$ -algebra generated by U . Projections are thus mapped onto projections and by comparing the left and right hand side in Eq.(6.37) we get $T(P_{k+1}) = P_k$ from the uniqueness of the spectral decomposition. $\square$

The last item implies that $T^m(P_k) = P_k$, i.e., if $m > 1$, then T^m is reducible. Similarly, if $l, m > 1$, then T^l is reducible if l and m have a non-trivial integer factor in common.

In the proof of the above theorem Eq.(6.36) is remarkable and has consequences beyond the peripheral spectrum: if the peripheral spectrum is non-trivial ($m > 1$), then every eigenvalue of T is an element of a group of eigenvalues of the same magnitude but rotated in the complex plane:

Proposition 6.7 (Covariance and eigensystem of irreducible maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be an irreducible positive and unital linear map which fulfills the Schwarz inequality. Following Thm.6.6 denote by $\{\gamma^k\}_{k=1,\dots,m}$ the peripheral spectrum of T and by $U = \bar{\gamma}T(U)$ the respective unitary eigenvector.*

1. For every $X \in \mathcal{M}_d(\mathbb{C})$ we have $T(U^\dagger X U) = U^\dagger T(X) U$.
2. If X is an eigenvector with $T(X) = \mu X$, then

$$T(U^\dagger X) = \bar{\gamma} \mu U^\dagger X, \quad T(X U) = \gamma \mu X U. \quad (6.38)$$

PROOF Both statements follow directly from Eq.(6.36) and its adjoint. $\square$

Note that the statements 1. and 2. taken together imply that if μ is non-degenerate, then $U^\dagger X U = e^{i\varphi} X$. So either $[X, U] = 0$ or $e^{i\varphi}$ is a root of unity and $\text{spec}(X) = e^{i\varphi} \text{spec}(X)$.

6.3 Primitive maps

We will continue the discussion on Perron-Frobenius theory and direct the focus to *primitive maps*. A primitive map is an irreducible positive map with trivial peripheral spectrum. Similar to the irreducible case primitive maps can, however, be characterized by a number of equivalent properties each of which may be used as a definition:

Theorem 6.7 (Primitive maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace-preserving linear map. Then the following statements are equivalent:*

1. *There exists an $n \in \mathbb{N}$ such that for every density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ we have $T^n(\rho) > 0$.*
2. *T^k is irreducible for every $k \in \mathbb{N}$ and the limit T_ϕ (see Prop.6.3) is irreducible as well.*
3. *For all density matrices ρ the limit $\lim_{k \rightarrow \infty} T^k(\rho)$ exists, is independent of ρ and given by a positive definite density matrix ρ_∞ .*
4. *T has trivial peripheral spectrum (i.e., counting algebraic multiplicities, there is only one eigenvalue of magnitude one) and the corresponding eigenvector is positive definite.*

PROOF 1. $\rightarrow$ 2. can be proven by contradiction: assume for instance $T_\phi = \lim_{i \rightarrow \infty} T^{n_i}$ is reducible. Then there is a density matrix ρ_0 which has a kernel and for which $\rho_0 = T_\phi(\rho_0) = \lim_i T^{n_i}(T^{n_i-n}(\rho))$ which would imply that $T^n(\rho) \not> 0$ for some ρ . Similarly, if T^k and therefore also T^{km} is reducible for all $m \in \mathbb{N}$, then there cannot be any n for which 1. holds.

2. $\rightarrow$ 4.: If T_ϕ has a non-degenerate eigenvalue one, then, by definition of T_ϕ , the map T has trivial peripheral spectrum. Moreover, since the fixed point set of T is included in the one of T_ϕ which in turn contains only a single positive definite element, this fixed point has to be the one of T as well.

4. $\rightarrow$ 3.: Since T has only one eigenvalue of magnitude one with corresponding positive definite eigenvector $\rho_\infty = T(\rho_\infty)$ we have $\lim_{k \rightarrow \infty} T^k = T_\phi$ and $T_\phi(\rho) = \rho_\infty \text{tr}[\rho]$ is a projection.

3. $\rightarrow$ 1.: First note that 3. implies that the eigenvalue of T which is second largest in magnitude, denote it by λ_2 , satisfies $|\lambda_2| < 1$. Assume that $T^n(\rho)$ would have a kernel with eigenvector ψ . Then

$$\lambda_{\min}(\rho_\infty) \leq |\langle \psi | T^n(\rho) - \rho_\infty | \psi \rangle| \leq \|T^n(\rho) - \rho_\infty\| \quad (6.39)$$

$$= \|(T^n - T_\infty)(\rho - \rho_\infty)\| \leq \mu^n c \|\rho - \rho_\infty\|, \quad (6.40)$$

where $\lambda_{\min}$ denotes the smallest eigenvalue, μ is such that $1 > \mu > |\lambda_2|$ and $\|\cdot\|$ is the operator norm. The constant $c > 0$ depends on T but is independent of n (see the discussion around Thm.8.23 for more details). Thus if n is sufficiently large, then Eqs.(6.39,6.40) would lead to a contradiction for every ρ . Hence, there is a finite n such that $T^n(\rho) > 0$ for all ρ .

□

If we interpret $T^t(\rho)$ for $t \in \mathbb{N}$ as discrete time-evolution, then primitive maps are exactly those for which the evolution eventually converges to a stationary state which is strictly positive (i.e., it may in some contexts be interpreted as a finite temperature Gibbs state) and independent of the initial state.

Note that if T in addition satisfies the Schwarz inequality as required by Thm.6.6, then we can replace point 2. in Thm.6.7 by the statement that T^k is irreducible for all $k < d^2$.

Moreover, if a map is completely positive, we can give three more equivalent characterizations of primitivity:

Theorem 6.8 (Completely positive primitive maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a completely positive and trace-preserving linear map with Kraus decomposition $T(\cdot) = \sum_i K_i \cdot K_i^\dagger$. Denote by $\mathcal{K}_m := \text{span}\{\prod_{k=1}^m K_{i_k}\}$ the complex linear span of all degree- m monomials of Kraus operators. Then the following are equivalent:*

1. T is primitive.
2. There exists an $n \in \mathbb{N}$ such that for every non-zero $\psi \in \mathbb{C}^d$ and all $m \geq n$: $\mathcal{K}_m|\psi\rangle = \mathbb{C}^d$.
3. There exists a $q \in \mathbb{N}$ such that for all $m \geq q$: $\mathcal{K}_m = \mathcal{M}_d(\mathbb{C})$.
4. There exists a $q \in \mathbb{N}$ such that for all $m \geq q$: $(T^m \otimes \text{id}_d)(|\Omega\rangle\langle\Omega|) > 0$, where $\Omega \in \mathbb{C}^d \otimes \mathbb{C}^d$ denotes a maximally entangled state.

The numbers q appearing in 3., 4. can be chosen equal and if q and n are chosen minimal, then $n \leq q$.

PROOF First consider item 2. and observe that it obviously holds for all $m \geq n$ if it holds for some n since we can always incorporate $m - n$ Kraus operators in ψ . Similar holds for item 4. In order to see this, define $\tau_q := (T^q \otimes \text{id}_d)(|\Omega\rangle\langle\Omega|)$ and assume $\tau_q > 0$. Then there is in particular a constant $c > 0$ such that $\tau_q \geq c\tau_{q-1}$ and therefore

$$\tau_{q+1} = (T \otimes \text{id})(\tau_q) \geq c(T \otimes \text{id})(\tau_{q-1}) = c\tau_q > 0. \quad (6.41)$$

By induction the same holds then true for all integers larger than q .

1. $\leftrightarrow$ 2.: Let us introduce $K_I := \prod_{k=1}^n K_{i_k}$ with multi-index $I = (i_1, \dots, i_n)$. If T is primitive, then by Thm.6.7 there exists an n such that $\sum_I K_I |\psi\rangle\langle\psi| K_I^\dagger$ has full rank for every ψ . Thus, for every ϕ there is an I so that $\langle\phi|K_I|\psi\rangle \neq 0$ and therefore $\text{span}\{K_I|\psi\rangle\} = \mathbb{C}^d$. Conversely, if the latter is true, then for every ϕ there will be a unitary U so that $|\phi\rangle \propto \sum_I U_{JI} K_I |\psi\rangle =: K'_J |\psi\rangle$. Since $\{K_I\}$ and $\{K'_J\}$ both provide a Kraus representation of T^n (see Thm.2.1) the full rank property is guaranteed. Following Thm.6.7 T then has to be primitive.

3. $\leftrightarrow$ 4.: Following the above lines and denoting by K_I the Kraus operators of T^q it is clear that $(T^q \otimes \text{id}_d)(|\Omega\rangle\langle\Omega|) > 0$ is equivalent to $\text{span}\{(K_I \otimes \mathbb{1}_d)|\Omega\rangle\} =$

$\mathbb{C}^d \otimes \mathbb{C}^d$. From here the equivalence between item 3. and 4. follows due to the fact that the relation $K \leftrightarrow (K \otimes \mathbb{1})|\Omega\rangle$ is a linear one-to-one mapping between $\mathcal{M}_d(\mathbb{C})$ and $\mathbb{C}^d \otimes \mathbb{C}^d$.

4. $\rightarrow$ 1.: If the Choi-Jamiolkowski matrix $\tau := (T^q \otimes \text{id}_d)(|\Omega\rangle\langle\Omega|)$ has full rank, then so does

$$T^q(|\psi\rangle\langle\psi|) = d(\mathbb{1}_d \otimes \langle\bar{\psi}|) \tau (\mathbb{1}_d \otimes |\bar{\psi}\rangle) \quad \text{for all } \psi \in \mathbb{C}^d. \quad (6.42)$$

This also shows that $n = q$ is sufficient. So if we choose both n and q minimal, then $n \leq q$ in general.

1. $\rightarrow$ 4.: The proof for this implication is entirely parallel with Eqs.(6.39,6.40) where q now plays the role of n : just replace ρ by Ω , ρ_∞ by $\rho_\infty \otimes \mathbb{1}/d$ and T by $T \otimes \text{id}$. Also the constant c now depends on $T \otimes \text{id}$, but the argumentation remains the same. $\square$

Note that all the above relations for both positive and completely positive maps hold as well for maps which are not trace-preserving³ so that the notion of primitivity is well-defined for arbitrary positive maps.

Finally, we will show that the number q (and therefore also n) appearing in Thm.6.8 can be bounded by the dimension d of the system. These bounds can be seen as a quantum counterpart to Wielandt's inequality, which does the same for the classical case of primitive stochastic matrices. We need two preparatory Lemmas before the main result. As before $\mathcal{K}_n$ denotes the complex linear span of the Kraus operators of T^n . We will frequently use that every $K \in \mathcal{K}_n$ is, up to a scalar multiple, a possible Kraus operator of T^n . This follows from the freedom in the Kraus decomposition (Thm.2.1, point 4.).

Lemma 6.2 *Let T be a primitive quantum channel on $\mathcal{M}_d(\mathbb{C})$ with Kraus rank k . Then, there is an natural number $n \leq d^2 - k + 1$ and a $K \in \mathcal{K}_n$ such that $\text{tr}[K] \neq 0$.*

PROOF Let us denote by R_n the complex linear span of all $\mathcal{K}_m$ with $m \leq n$. We have to show that: (*) for any $n \in \mathbb{N}$, if $\dim[R_n] < d^2$, then $\dim[R_{n+1}] > \dim[R_n]$. Since $\dim[R_1] = k$, by iteration we obtain that $R_{d^2-k+1} = \mathcal{M}_d(\mathbb{C})$. This implies that a linear combination of the elements of $\mathcal{K}_n$ with various $n \leq d^2 - k + 1$ must be equal to the identity, and thus at least one of the elements must have non-zero trace. To prove (*) we note that, by definition, $R_n \subseteq R_{n+1}$. If they would be equal, then $R_m = R_n$ for all $m > n$. Thus, equality can only occur when $\dim[R_n] = d^2$ since otherwise the map T would not be primitive.

Lemma 6.3 *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a primitive completely positive map such that at least one of its Kraus operators, say K_1 , has a non-zero eigenvalue, i.e., $K_1|\varphi\rangle = \mu|\varphi\rangle$ with $\mu \neq 0$. Then: (a) $\mathcal{K}_{d-1}|\varphi\rangle = \mathbb{C}^d$. (b) If K_1 is not invertible, then for all $|\psi\rangle \in \mathbb{C}^d$, $|\varphi\rangle\langle\psi| \in \mathcal{K}_{d^2-d+1}$.*

³The limit $\lim_{k \rightarrow \infty} T^k$ has then to be normalized by the k 'th power of the spectral radius of T .

PROOF (a) We define S_n as the span of all $\mathcal{K}_m|\varphi\rangle$ with $m \leq n$ together with $|\varphi\rangle$. If $\dim[S_n] < d$, then $\dim[S_{n+1}] > \dim[S_n]$, since otherwise the map would not be primitive. Thus, $S_{d-1} = \mathbb{C}^d$. That is, for all $|\phi\rangle \in \mathbb{C}^d$, there exist matrices $K^{(n)} \in \mathcal{K}_{k_n}$, $k_n \leq d-1$ such that (with $K^{(0)} \propto \mathbb{1}$)

$$|\phi\rangle = \sum_{n=0}^{d-1} K^{(n)}|\varphi\rangle = \sum_{n=0}^{d-1} K^{(n)} \frac{K_1^{d-k_n}}{\mu^{d-k_n}}|\varphi\rangle, \quad (6.43)$$

and thus, $|\phi\rangle \in \mathcal{K}_{d-1}|\varphi\rangle$. (b) We write K_1 in the Jordan standard form and divide it into two blocks. The first one, of size $\tilde{d} \times \tilde{d}$, consists of all Jordan blocks corresponding to non-zero eigenvalues, whereas the second one contains all those corresponding to zero eigenvalues. We denote by P the (not necessarily Hermitian) projector onto the subspace where the first block is supported and by $r \leq d - \tilde{d}$ the size of the largest Jordan block corresponding to a zero eigenvalue. We have

$$K_1 P = P K_1, \quad K_1^r = K_1^r P. \quad (6.44)$$

We define $R_n := P K_n$ and show that $R_{d\tilde{d}} = P \mathcal{M}_d(\mathbb{C})$. For all $n \in \mathbb{N}$, $\dim[R_{n+1}] \geq \dim[R_n]$. The reason is that for any set of linearly independent matrices $K_k^{(n)} \in R_n$, $K_1 K_k^{(n)} \in R_{n+1}$ are also linearly independent, given that K_1 is invertible on its range. By following the reasoning of [?, Appendix A] we get that, if $\dim[R_{n+1}] = \dim[R_n] =: d'$, then $\dim[R_m] = d'$ for all $m > n$, which is incompatible with T being primitive unless $d' = \tilde{d}d$. Thus, for any $|\psi\rangle \in \mathbb{C}^d$, there exists a $K \in \mathcal{K}_{\tilde{d}d}$ with $|\varphi\rangle\langle\psi| = PK = K_1^r P K / \mu^r = K_1^r K / \mu^r \in \mathcal{K}_{\tilde{d}d+r}$. By using that $\tilde{d} \leq d-r$ and that $r \geq 1$ (since K_1 is supposed to be not invertible) we get $\tilde{d}d + r \leq d^2 - d + 1$, which concludes the proof.

We have now the necessary tools to prove our main result.

Theorem 6.9 (Quantum Wielandt inequality) *Let T be a primitive quantum channel on $\mathcal{M}_d(\mathbb{C})$ with Kraus rank k . Then for the minimal q as it appears in Thm.6.8 we have that*

1. in general $q \leq (d^2 - k + 1)d^2$,
2. if the span of Kraus operators $\mathcal{K}_1$ contains an invertible element, then $q \leq d^2 - k + 1$,
3. if the span of Kraus operators $\mathcal{K}_1$ contains an element with at least one non-zero eigenvalue, then $q \leq d^2$.

PROOF

2. If there is an invertible element, then it follows from [?, Appendix A, Proposition 2] that $\dim \mathcal{K}_{n+1} > \dim \mathcal{K}_n$ until the full matrix space $\mathcal{M}_d(\mathbb{C})$ is spanned and thus $q \leq d^2 - k + 1$.

1. Let us denote by $\{K_k^{(n)}\}$ the Kraus operators corresponding to T^n . According to Lemma 6.2, one of them, say $K_1^{(n)}$, has non-zero trace for some $n \leq d^2 - k + 1$ and therefore there exists $|\varphi\rangle$ such that $K_1^{(n)}|\varphi\rangle = \mu|\varphi\rangle$ with

$\mu \neq 0$. If $K_1^{(n)}$ is invertible, then 1. is implied by 2., so we can assume that it is not invertible. According to Lemma 6.3.(b), for all $|\psi\rangle, |\chi\rangle \in \mathbb{C}^d$ we have $|\varphi\rangle\langle\psi| \in \mathcal{K}_{(d^2-d+1)n}$; and according to Lemma 6.3.(a) $|\chi\rangle\langle\psi| \in \mathcal{K}_{nd^2}$. This implies that $\mathcal{K}_{nd^2} = \mathcal{M}_d(\mathbb{C})$ and hence the general bound 1. follows.

The argument which proves 3. is completely analogous just with $n = 1$ as we do, by assumption, not have to block in order to get a Kraus operator with non-zero eigenvalue. $\square$

In the previous proof we use, for the general case, blocks of T^n for some $n \leq d^2 - k + 1$ just in order to get any Kraus operator with a non-zero eigenvalue. This rather clumsy step is not necessary if the dimension is small:

Corollary 6.4 *Let T be a primitive quantum channel on $\mathcal{M}_d(\mathbb{C})$ with $d = 2, 3$. Then there is always a Kraus operator with non-zero eigenvalue in $\mathcal{K}_1$ and thus $q \leq d^2$.*

PROOF The fact that $\mathcal{K}_1$ has this property for $d = 2, 3$ stems from the classification of nilpotent subspaces[?]: assume that $\mathcal{K}_1$ would be a nilpotent subspace within the space of $d \times d$ matrices. Then for $d = 2$ its dimension would have to be one, so it could not arise from the Kraus operators of a quantum channel. Similarly, for $d = 3$ there are (up to similarity transformations) two types of nilpotent subspaces [?] of dimension $k > 1$: one of dimension $k = 3$, the space of upper-triangular matrices, whose structure does not allow the trace-preserving property, and one of dimension $k = 2$ which only leads to quantum channels having a (in modulus) degenerate largest eigenvalue. Hence, if $\mathcal{K}_1$ is generated by the Kraus operators of a primitive quantum channel, then it cannot be nilpotent if $d = 2, 3$. $\square$

For $d \geq 4$ the set of Kraus operators can form a nilpotent subspace even for primitive quantum channels. However, exploiting the structure of such subspaces[?] the general bound in Thm.6.9 can still be improved.

Problem 12 (Optimal Wielandt bound) *Find the smallest integers n and q (as a function of the dimension d) for Thm.6.8.*

6.4 Fixed points

Theorem 6.10 (Brouwer's fixed point theorem) *Let T be a continuous map from a non-empty, compact, convex set $S \subset \mathbb{R}^n$ into itself, then there is an $x \in S$ such that $T(x) = x$.*

As the set of Hermitian matrices in $\mathcal{M}_d(\mathbb{C})$ is a real vector space and the density matrices therein form a compact, convex set, Brouwer's fixed point theorem immediately implies:

Theorem 6.11 (Stationary states) *Every continuous, trace-preserving, positive (not necessarily linear) map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ has at least one stationary state. That is, there is a density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ such that $T(\rho) = \rho$.*

In the following we are interested in linear maps, which are automatically continuous.

Proposition 6.8 (Positive fixed-points) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map and $X = T(X)$ a fixed point. If $X = \sum_{j=1}^4 c_j P_j$ is the decomposition of X into four positive operators $P_j \geq 0$ which arises from decomposing X first into Hermitian and anti-Hermitian parts and these further into orthogonal positive and negative parts, then*

$$T(P_j) = P_j \quad j = 1, \dots, 4, \quad \text{are fixed points as well.} \quad (6.45)$$

PROOF For every Hermiticity preserving linear map we obviously have that the two Hermitian operators $X + X^\dagger$ and $i(X - X^\dagger)$ are fixed points if X is one. So let us assume that X is Hermitian and that $P_\pm \geq 0$ are its orthogonal positive and negative parts, i.e., $X = P_+ - P_-$ and $\text{tr}[P_+ P_-] = 0$. The fixed point equation then reads $T(P_+) - T(P_-) = P_+ - P_-$ and we want to identify $T(P_\pm)$ with $P_\pm$. To this end, denote by Q the projection onto the support of P_+ and note that the fixed point equation together with the trace-preserving property of T implies that

$$\text{tr}[P_+] = \text{tr}[Q(P_+ - P_-)] = \text{tr}[Q(T(P_+ - P_-))] \quad (6.46)$$

$$\leq \text{tr}[T(P_+)] = \text{tr}[P_+]. \quad (6.47)$$

Since equality has to hold for the inequality in Eq.(6.47), we have to have that $T(P_+)$ is supported on Q while $T(P_-)$ is orthogonal to it. Consequently, T preserves the orthogonality of $P_\pm$ and we have $T(P_\pm) = P_\pm$. $\square$

For a map T on $\mathcal{M}_d(\mathbb{C})$ let us define the set of fixed points

$$\mathcal{F}_T := \{X \in \mathcal{M}_d(\mathbb{C}) \mid X = T(X)\}. \quad (6.48)$$

Note that if T is linear, then $\mathcal{F}_T$ is closed under linear combination, i.e., it is a vector space, and if T is Hermiticity preserving (in particular, if it is positive), then $\mathcal{F}$ is closed under Hermitian conjugation. The previous Prop.6.8 immediately implies that for trace-preserving quantum channels the set of fixed points is spanned by stationary density operators:

Corollary 6.5 (Linearly independent stationary states) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map. If the space $\mathcal{F}_T$ has dimension D , then there are D linearly independent density operators $\{\rho_\alpha \in \mathcal{M}_d(\mathbb{C})\}_{\alpha=1,\dots,D}$ such that $T(\rho_\alpha) = \rho_\alpha$ and*

$$\mathcal{F}_T = \text{span}\{\rho_1, \dots, \rho_D\}. \quad (6.49)$$

Proposition 6.9 (Maximal support of fixed points) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map and T_∞ the projection onto its fixed point space as given in Eq.(6.14). Then every fixed point $X = T(X) \in \mathcal{M}_d(\mathbb{C})$ has support and range within the support space of the fixed point $T_\infty(\mathbb{1})$.*

PROOF For positive X the claim follows from positivity of T_∞ : since $0 \leq T_\infty(X) = X \leq \|X\|_\infty T_\infty(\mathbb{1})$ the range (which equals the support) of X has to be contained in the one of $T_\infty(\mathbb{1})$ (See Douglas' theorem 5.1).

If X is not positive we can by Prop.6.8 decompose it into a linear combination of four positive fixed points $P_1, \dots, P_4$. Using that $\text{supp}(X)$ is in the linear span of $\bigcup_j \text{supp}(P_j)$ together with the result for positive fixed points then shows $\text{supp}(X) \subseteq \text{supp}(T_\infty(\mathbb{1}))$. The same argument applies for the range of X . Alternatively, we may use that X is a fixed point iff $X^\dagger$ is, and that the range of X equals the support of $X^\dagger$. $\square$

The following shows that the ranges of all stationary states play an exceptional role: every state which is supported within such a space will remain so after the action of the channel:

Proposition 6.10 (Stationary subspaces) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map, and let $Q \in \mathcal{M}_d(\mathbb{C})$ be the Hermitian projection onto the support of any stationary density operator $\sigma = T(\sigma) \in \mathcal{M}_d(\mathbb{C})$. Then $\text{tr}[(\mathbb{1} - Q)T(Q)] = 0$ and for every density operator $\rho \in \mathcal{M}_d(\mathbb{C})$*

$$\rho \leq Q \quad \Rightarrow \quad T(\rho) \leq Q. \quad (6.50)$$

PROOF By definition of Q we have that $Q \leq c\sigma \leq c'Q$ for some strictly positive numbers c and c' . Therefore $T(Q) \leq cT(\sigma) = c\sigma \leq c'Q$ which implies $\text{tr}[(\mathbb{1} - Q)T(Q)] = 0$. Similarly, if $\rho \leq Q$, then $T(\rho) \leq c'Q$, i.e., $T(\rho)$ has support in the range of Q . Since $T(\rho)$ has eigenvalues at most one, this means that $T(\rho) \leq Q$. $\square$

Projections onto stationary subspaces (in the sense of Eq.(6.50)) can be easily characterized in the Heisenberg picture, as well:

Proposition 6.11 (Stationary subspaces II) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map, and let $Q \in \mathcal{M}_d(\mathbb{C})$ be any Hermitian projection. Then the following statements are equivalent:*

1. *For every density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ with $\rho \leq Q$, we have that $T(\rho) \leq Q$ as well,*
2. *$T^*(Q) \geq Q$.*

PROOF 1. $\rightarrow$ 2.: define the orthogonal projection $P := (\mathbb{1} - Q)$. The assumption implies that $QT^*(P)Q = 0$ which in turn means that $PT^*(P)P = T^*(P)$. Using that $T^*(\mathbb{1}) = \mathbb{1}$, the latter can be rewritten as $T^*(Q) - PT^*(Q)P = Q$ from which $T^*(Q) \geq Q$ follows. 2. $\rightarrow$ 1. follows from the chain of inequalities

$$0 \leq \text{tr}[T(\rho)P] = 1 - \text{tr}[\rho T^*(Q)] \leq 1 - \text{tr}[\rho Q] = 0.$$

□

A projection onto a stationary subspace is often called *sub-harmonic* and item 2. of the previous proposition is used as the defining property of this notion. Note that while the support projection of every stationary state is sub-harmonic (due to Props.6.10,6.11), the converse is not necessarily true: for instance $T^*(\mathbb{1}) \geq \mathbb{1}$ holds trivially for every trace-preserving T even if there is no stationary state of full rank.

If T is completely positive with Kraus decomposition $T(\cdot) = \sum_i K_i \cdot K_i^\dagger$, one can relate support spaces of fixed points to properties of the Kraus operators: suppose there is a stationary state with support $\mathcal{H} \in \mathbb{C}^d$. Then all Kraus operators have to preserve this subspace in the sense that $K_i \mathcal{H} \subseteq \mathcal{H}$. In particular, if $\rho = |\psi\rangle\langle\psi|$ is a pure state, then ψ has to be a simultaneous eigenvector of all Kraus operators.

We will now investigate the structure of the set of fixed points $\mathcal{F}$ of quantum channels. As a first step we exploit Props.6.9,6.10 in order to restrict the discussion to trace-preserving maps which have a full rank fixed point:

Lemma 6.4 (Restriction to full rank fixed points) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, (completely) positive, linear map, and $\mathcal{H} \subseteq \mathbb{C}^d$ the support space of $T_\infty(\mathbb{1})$. Then the linear map $\tilde{T} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ defined via $\tilde{T}(X) := T(X \oplus 0)|_{\mathcal{H}}$ is trace-preserving and (completely) positive as well. Moreover, $\tilde{T}$ has a fixed point of full rank and⁴*

$$\mathcal{F}_T = \mathcal{F}_{\tilde{T}} \oplus 0. \quad (6.51)$$

PROOF Denote by $V : \mathcal{H} \rightarrow \mathbb{C}^d$ the isometry which embeds $\mathcal{H}$ into $\mathbb{C}^d$ so that $V^\dagger V = \mathbb{1}$ and $Q := VV^\dagger$ is the projection onto the range of $T_\infty(\mathbb{1})$. Then by definition $\tilde{T}(X) = V^\dagger T(VXV^\dagger)V$ so that $\tilde{T}$ inherits the property of being (completely) positive from T . In order to see that $\tilde{T}$ is trace-preserving note that $VXV^\dagger = X \oplus 0$ has support and range within the range of Q . Thus we can decompose it into four positive operators with the same property and apply Prop.6.10 which leads to

$$T(X \oplus 0) = \tilde{T}(X) \oplus 0, \quad (6.52)$$

so that the trace-preserving property is inherited from T as well. Finally, Eq.(6.52) also implies Eq.(6.51) since by Prop.6.9 all fixed points of T are of the form $X \oplus 0$. In particular, by definition of $\mathcal{H}$, there is one fixed point which has full rank on $\mathcal{H}$. □

⁴Here the ‘0’ is supposed to act on the orthogonal complement of $\mathcal{H}$ in $\mathbb{C}^d$.

Before returning to trace-preserving maps we will discuss their duals–unital maps. This will provide an important prerequisite for the discussion of the trace-preserving case. We will also see that the structure of the fixed point set of a map simplifies a lot when the dual map has at least one fixed point of full rank. This is already visible in the dual analogue of Lem.6.4:

Lemma 6.5 *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map, and $\tilde{T}$ and V defined as in Lem.6.4. Let us further denote by $\mathcal{S}$ the set of linear unital maps which satisfy the Schwarz inequality in Eq.(5.2). Then*

$$A \in \mathcal{F}_{T^*} \Rightarrow V^\dagger A V \in \mathcal{F}_{\tilde{T}^*} \quad \text{and} \quad (6.53)$$

$$X \in \mathcal{F}_{\tilde{T}^*} \Leftrightarrow V^\dagger T^*(V X V^\dagger) V = X. \quad (6.54)$$

$$T^* \in \mathcal{S} \Rightarrow \tilde{T}^* \in \mathcal{S}. \quad (6.55)$$

PROOF Eq.(6.54) follows from inserting the definitions. For Eq.(6.53) note that $\tilde{T}^*(V^\dagger A V) = V^\dagger T^*(Q A Q) V$ (using the notation of Lem.6.4 again). Taking the trace with an arbitrary $C \in \mathcal{B}(\mathcal{H})$ we obtain

$$\text{tr} [C \tilde{T}^*(V^\dagger A V)] = \text{tr} [T(V C V^\dagger) Q A Q] \quad (6.56)$$

$$= \text{tr} [T(V C V^\dagger) A] = \text{tr} [C V^\dagger A V], \quad (6.57)$$

where the second equality follows from Prop.6.10 and the last step uses the assumption that A is a fixed point of T^* . Finally, the Schwarz inequality for $\tilde{T}^*$ follows from

$$V^\dagger T^*(V A^\dagger A V^\dagger) V \geq V^\dagger T^*(V A^\dagger V^\dagger) T^*(V A V^\dagger) V \quad (6.58)$$

$$\geq V^\dagger T^*(V A^\dagger V^\dagger) V V^\dagger T^*(V A V^\dagger) V. \quad (6.59)$$

Here the first inequality comes from first inserting $V^\dagger V = \mathbb{1}$ between $A^\dagger$ and A and then using the Schwarz inequality for T^* and the second inequality is due to $V V^\dagger \leq \mathbb{1}$. Written in terms of $\tilde{T}^*$ the above equations lead to the claimed $\tilde{T}^*(A^\dagger A) \geq \tilde{T}^*(A^\dagger) \tilde{T}^*(A)$. $\square$

Due to the aforementioned issue we will first characterize the fixed point set of unital maps which have a dual with fixed point of full rank:

Theorem 6.12 (Structure of fixed points for unital Schwarz maps) *Let $T^* : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a unital map which (i) satisfies the Schwarz inequality Eq.(5.2) (e.g., a completely positive map) and (ii) is such that T has a full rank fixed point. Then the set of fixed points $\mathcal{F}_{T^*}$ is a $*$ -algebra and has thus the form specified in Eq.(1.39) (without the zero-block).*

PROOF Let $\rho \in \mathcal{M}_d(\mathbb{C})$ be a full rank fixed point of T . Note that by Prop.6.9 we can w.l.o.g. chose a positive definite $\rho > 0$. Denote by $A = T^*(\rho) \in \mathcal{M}_d(\mathbb{C})$ a fixed point of T^* . Then, using the fixed point properties and the fact that $A^\dagger$ is a fixed point of T^* as well we obtain

$$\text{tr} [(T^*(A^\dagger A) - T^*(A^\dagger) T^*(A)) \rho] = 0. \quad (6.60)$$

Note that the expression in parentheses is positive semi-definite by the Schwarz inequality Eq.(5.2) and therefore, in fact, identical zero since ρ is assumed to be positive definite. Hence, we have equality in the Schwarz inequality so that we can apply Thm.5.4 which implies that if $A, X \in \mathcal{F}_{T^*}$, then $AX \in \mathcal{F}_{T^*}$ is a fixed point as well. Since $\mathcal{F}_{T^*}$ is also closed under linear combinations and Hermitian conjugation, it is a $*$ -algebra. Evidently, it is unital since $T^*(1) = 1$ holds by assumption. $\square$

If T does not have a full rank fixed point, then $\mathcal{F}_{T^*}$ need not be an algebra. Together with Lem.6.5 we obtain a weaker structure in this case:

Corollary 6.6 *Let $T^* : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a unital linear map which satisfies the Schwarz inequality Eq.(5.2). Let $Q = Q^\dagger \in \mathcal{M}_d(\mathbb{C})$ be the projection onto the maximum rank fixed point of T , then the following set is a $*$ -algebra:*

$$\{Y \in Q\mathcal{M}_d(\mathbb{C})Q \mid QT^*(Y)Q = Y\}. \quad (6.61)$$

PROOF The stated set is exactly the one appearing in Eq.(6.54), i.e., the fixed point set of $\tilde{T}^*$. Since $\tilde{T}$ has, however, a fixed point of full rank by construction (see Lem.6.4) we can apply Thm.6.12 which proves the claim. $\square$

The relation between fixed point sets of unital maps and $*$ -algebras becomes more transparent for completely positive maps:

Theorem 6.13 (Fixed points and commutants) *Let $T^* : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a completely positive unital map with Kraus decomposition $T^*(\cdot) = \sum_i K_i^\dagger \cdot K_i$. If both $X, X^\dagger X \in \mathcal{F}_{T^*}$, then $[X, K_i] = 0$ for all i . Consequently, the largest $*$ -subalgebra contained within $\mathcal{F}_{T^*}$ is given by the set*

$$\{X \in \mathcal{M}_d(\mathbb{C}) \mid \forall i : [X, K_i] = [X, K_i^\dagger] = 0\}. \quad (6.62)$$

PROOF The assertion follows from the equation

$$\sum_i [X, K_i]^\dagger [X, K_i] = T^*(X^\dagger X) - X^\dagger T^*(X) - T^*(X)^\dagger X + X^\dagger X = 0$$

by exploiting the fixed point properties of X and $X^\dagger X$ together with the fact that the l.h.s. is a sum over positive terms which is zero iff all terms vanish individually. Hence every $*$ -subalgebra of $\mathcal{F}_{T^*}$ is a subset of Eq.(6.62). Conversely, the set in Eq.(6.62) is a $*$ -subalgebra within $\mathcal{F}_{T^*}$ on its own and thus the largest one. $\square$

Finally, we discuss the structure of the fixed point set of trace-preserving maps:

Theorem 6.14 (Fixed points of trace-preserving maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map for which T^* satisfies the Schwarz inequality in Eq.(5.2). Then there is a unitary $U \in \mathcal{M}_d(\mathbb{C})$ and a set of positive definite density matrices $\rho_k \in \mathcal{M}_{m_k}(\mathbb{C})$ such that the fixed point set of T is given by*

$$\mathcal{F}_T = U \left(0 \oplus \bigoplus_{k=1}^K \mathcal{M}_{d_k} \otimes \rho_k \right) U^\dagger, \quad (6.63)$$

for an appropriate decomposition of the Hilbert space $\mathbb{C}^d = \mathbb{C}^{d_0} \oplus \bigoplus_k \mathbb{C}^{d_k} \otimes \mathbb{C}^{m_k}$. $\mathcal{M}_{d_k}$ stands for the full algebra of complex matrices on $\mathbb{C}^{d_k}$.

PROOF We exploit that by Lem.6.4 $\mathcal{F}_T = 0 \oplus \mathcal{F}_{\tilde{T}}$ after an appropriate basis transformation which we can assign to a unitary U . Now $\tilde{T}$ has, by construction, a full rank fixed point and following Lem.6.4 it is trace-preserving and its dual satisfies the Schwarz inequality due to Lem.6.5. This implies by Thm.6.12 that $\tilde{T}_\infty^*$ projects onto a $*$ -algebra. The general form of a positive map projecting onto a $*$ -algebra (i.e., a conditional expectation) is given in Eq.(1.40). Taking the dual of this map we arrive at $\tilde{T}_\infty$ projecting onto a set of the form in Eq.(6.63) (without the zero-block). Putting things together and using that we can always move the kernel of each ρ_k into the zero-block completes the proof. $\square$

Note that Thm.6.14 implies that the fixed point $T_\infty(\mathbb{1})$ is (up to normalization) given by $U(0 \oplus \bigoplus_k \mathbb{1}_{m_k} \otimes \rho_k)U^\dagger$. Moreover, it shows that again the set of fixed points is closely related to a $*$ -algebra (actually is a $*$ -algebra w.r.t. a modified product):

Corollary 6.7 *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving, positive, linear map for which T^* satisfies the Schwarz inequality in Eq.(5.2). Then for any density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ which is a maximum rank fixed point of T we have that the set $\rho^{-1/2} \mathcal{F}_T \rho^{-1/2}$ is a $*$ -algebra (with the inverse taken on the support of ρ).*

Another interesting consequence of Thm.6.14 is its implication according to Prop.6.10: the decomposition of the fixed point space into a direct sum in Eq.(6.63) implies the same decomposition into stationary subspaces. What is more, the $\mathcal{M}_{d_k}$'s give rise to subsystem which are invariant, i.e., on which T acts as the identity (up to the reversible rotation given by U).

Theorem 6.15 (Unique fixed points of full rank) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving quantum channel with Kraus decomposition $T(\cdot) = \sum_i K_i \cdot K_i^\dagger$. If for some $n \in \mathbb{N}$ we have that*

$$\text{span} \left\{ \prod_{k=1}^n K_{i_k} \right\} = \mathcal{M}_d(\mathbb{C}), \quad (6.64)$$

i.e., homogeneous polynomials of the Kraus operators span the entire matrix algebra, then there exist a unique positive definite density matrix $\rho \in \mathcal{M}_d(\mathbb{C})$ such that $\mathcal{F}_T \propto \rho$.

PROOF Consider the map T^n , i.e., the n -fold concatenation of T . This has Kraus operators $R_I := \prod_{k=1}^n K_{i_k}$ with multi-index $I = (i_1, \dots, i_n)$ and it satisfies $\mathcal{F}_T \subseteq \mathcal{F}_{T^n}$. Now for every $\psi \in \mathbb{C}^d$ we have that $T^n(|\psi\rangle\langle\psi|)$ has full rank since by assumption $\text{span}\{R_I|\psi\rangle\} = \mathbb{C}^d$. This implies that there cannot be any stationary density operator which has a kernel. Since we can always construct a rank deficient positive fixed point if the dimension of $\mathcal{F}_T$ is larger than one (see Cor.6.5), the assertion follows. $\square$

Problem 13 (Fixed points of general positive maps) *Characterize the set of fixed points of unital, positive, linear maps which do not satisfy the Schwarz inequality.*

6.5 Cycles and recurrences

We will now have a closer look at the space corresponding to the peripheral spectrum (i.e., eigenvalues which are phases) of a positive linear map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ which is assumed to be either trace-preserving or unital. Denote the complex linear span of all the respective eigenspaces by

$$\mathcal{X}_T := \text{span}\{X \in \mathcal{M}_d(\mathbb{C}) \mid \exists \varphi \in \mathbb{R} : T(X) = e^{i\varphi} X\}, \quad (6.65)$$

for which we will simply write $\mathcal{X}$ if the dependence on T is clear from the context. The questions which we address first are: what's the structure of $\mathcal{X}$? what's the structure of T restricted to $\mathcal{X}$? and what does this imply on the peripheral spectrum?

Note that evidently $\mathcal{X}_T = \mathcal{X}_{T_\varphi}$ if T_φ is constructed from T as defined in Eq.(6.13) by discarding all but those terms in the spectral decomposition which correspond to the peripheral spectrum. Recall that by Prop.6.3 if T is trace-preserving or unital, then T_φ will be (completely) positive if T is and that by Prop.6.2 there cannot any non-trivial Jordan-block associated to an eigenvalue of modulus one.

Proposition 6.12 (Asymptotic image) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace-preserving linear map. Then $\mathcal{X}_T$ is the image of the projection T_ϕ , it is spanned by positive operators, and it is closed under the action of T . That is,*

1. $T_\phi(\mathcal{M}_d(\mathbb{C})) = \mathcal{X}_T$,
2. $\exists \{\rho_i \geq 0\} : \mathcal{X}_T = \text{span}\{\rho_i\}$,
3. $T(\mathcal{X}_T) = \mathcal{X}_T$.

PROOF Note that by the proof of Prop.6.3 every $X \in \mathcal{X}_T$ is a fixed point of T_ϕ since there is an ascending sequence $n_i \in \mathbb{N}$ such that $\lim_{i \rightarrow \infty} T_\phi^{n_i} = T_\phi$. Conversely, if $X = T_\phi(X)$ is such a fixed point, then it has to be a linear combination of eigenvectors of T which correspond to eigenvalues of modulus one. $\mathcal{X}_T$ thus coincides with the space of fixed points of T_ϕ which is in turn given by $T_\phi(\mathcal{M}_d(\mathbb{C}))$, proving 1.

2. follows from 1. together with Corr.6.5 and in order to arrive at 3. we just have to use the definition of $\mathcal{X}_T$ and express an element $X \in \mathcal{X}_T$ as a linear combination of peripheral eigenvectors of T . $\square$

The following refines this Proposition by adding the assumption that the map under consideration satisfies the Schwarz inequality. Exploiting this fact one observes that $\mathcal{X}$ is a $*$ -algebra (w.r.t. a modified product) and that T acts on it either by permuting blocks or by unitary conjugation on subsystems:

Theorem 6.16 (Structure of cycles) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a trace-preserving and positive linear map which fulfills the Schwarz inequality in Eq.(5.2).*

1. *There exists a decomposition of the Hilbert space $\mathbb{C}^d = \mathcal{H}_0 \oplus \bigoplus_{k=1}^K \mathcal{H}_k$ into a direct sum of tensor products⁵ $\mathcal{H}_k = \mathcal{H}_{k,1} \otimes \mathcal{H}_{k,2}$ and positive definite density matrices ρ_k acting on $\mathcal{H}_{k,2}$ such that*

$$\mathcal{X}_T = 0 \oplus \bigoplus_{k=1}^K \mathcal{M}_{d_k} \otimes \rho_k, \quad (6.66)$$

where $\mathcal{M}_{d_k}$ is a full complex matrix algebra on $\mathcal{H}_{k,1}$ of dimension $d_k := \dim(\mathcal{H}_{k,1})$. That is, for every $X \in \mathcal{X}_T$ there are $x_k \in \mathcal{B}(\mathcal{H}_{k,1})$ such that

$$X = 0 \oplus \bigoplus_k x_k \otimes \rho_k. \quad (6.67)$$

2. *There exist unitaries $U_k \in \mathcal{B}(\mathcal{H}_{k,1})$ and a permutation π , which permutes within subsets of $\{1, \dots, K\}$ for which the corresponding $\mathcal{H}_k$'s have equal dimension, so that for every $X \in \mathcal{X}_T$ represented as in Eq.(6.67)*

$$T(X) = 0 \oplus \bigoplus_{k=1}^K U_k x_{\pi(k)} U_k^\dagger \otimes \rho_k. \quad (6.68)$$

PROOF The basic ingredients of the proof are those which already appeared in the proof of Thm.6.1 where we characterized maps with unit determinant. Exploiting Dirichlet's lemma on Diophantine approximations (Lem.6.1) we can find an ascending subsequence $n_i \in \mathbb{N}$ so that $\lim_{i \rightarrow \infty} T^{n_i}$ converges to a map I with eigenvalues which are either one or zero. Clearly, I is again a trace-preserving, positive linear map which satisfies the Schwarz inequality since all these properties are preserved under concatenation. Moreover, $\mathcal{X}_T = \mathcal{X}_I$ is the fixed-point space of I . So, statement 1. of the theorem follows from the structure of the space of fixed points provided in Thm.6.14.

2. The map $T^{-1} := \lim_{i \rightarrow \infty} T^{n_i-1}$ is the inverse of T on $\mathcal{X}$ since by construction $T^{-1}T = I$. Hence, both T and T^{-1} map $\mathcal{X} \rightarrow \mathcal{X}$ in a bijective way. The crucial point here is that T^{-1} is again a positive, trace-preserving map since it is constructed as a limit of such maps. To understand the consequences, consider any pure state in $\mathcal{X}$, i.e., a density operator $\sigma \in \mathcal{X}$ which has no non-trivial convex decomposition within $\mathcal{X}$. Then the image of σ under T has to be a pure state as well: assume this is not the case, i.e., $T(\sigma) = \sum_i \lambda_i \sigma_i$ is a non-trivial convex decomposition into states $\sigma_i \in \mathcal{X}$. Then applying T^{-1} to this equation leads to a contradiction since $\sigma = \sum_i \lambda_i T^{-1}(\sigma_i)$ is not pure. Consequently, both T and T^{-1} map pure states in $\mathcal{X}$ onto pure states. Note that a

⁵Note that we may have $\dim \mathcal{H}_0 = 0$ as well as $\dim \mathcal{H}_{k,i} = 1$. Moreover, the direct sum in the decomposition does not necessarily correspond to a block structure in computational basis – just in *some* basis.

pure state can only have support in one of the K blocks, for instance $\sigma = x \otimes \rho_k$ where $x \in \mathcal{B}(\mathcal{H}_{k,1})$ is a rank one projection. Now we know that $T(\sigma) = x' \otimes \rho_{k'}$ for some k' and some rank one projection $x' \in \mathcal{B}(\mathcal{H}_{k',1})$. By continuity and the fact that T is a bijective linear map on $\mathcal{X}$, we have that within $\mathcal{X}$: (i) every element of block k is mapped to an element of the block k' (i.e., k' does not depend on x), and (ii) the blocks k and k' must have equal dimension $d_k = d_{k'}$. Therefore, there is a permutation π which permutes blocks of equal size so that $X \in \mathcal{X}$ is mapped to

$$T(X) = 0 \oplus \bigoplus_k T_k(x_{\pi(k)}) \oplus \rho_k,$$

with some linear maps $T_k : \mathcal{M}_{d_k} \rightarrow \mathcal{M}_{d_k}$. Since the latter have, together with their inverses, to be positive and trace-preserving they must be either matrix transpositions or unitary conjugations by Cor.6.2. Matrix transpositions are, however, ruled out by the requirement that T and thus each T_k is a Schwarz-map. $\square$

6.6 Inverse eigenvalue problems

6.7 Literature

There are various extensions of Brouwer's fixed point theorem. *Schauder's theorem*, for instance, extends it to topological vector spaces, which may be of infinite dimension. That is, S becomes a convex, compact subset of a topological vector space. *Kakutani's theorem* is a generalization to upper semi-continuous set-valued functions (correspondences) T from a non-empty, compact, convex set $S \in \mathbb{R}^n$ into itself such that for all $x \in S$ the set $T(x)$ is non-empty and convex, too. Dealing with these and more refined fixed point theorems is part of *algebraic topology*.